

# Dense Caption Reconstruction: Quantifying Causal Narrative Inference and Visual Decoupling in Video Language Models

Anonymous ICCV submission

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## Abstract

001 *Current Video-Language Models (Video-LLMs) routinely*  
002 *process dense visual frames as an uncompressed, uniform*  
003 *token sequence, incurring severe computational costs re-*  
004 *gardless of the video’s underlying information density. Yet,*  
005 *many real-world physical events follow structured causal*  
006 *scripts that can be inferred without continuous optical obser-*  
007 *vation. In this paper, we establish a diagnostic framework*  
008 *for **Dense Caption Reconstruction** to measure the boundary*  
009 *between **symbolic narrative inference** (what causally must*  
010 *occur) and **visual perception** (what optical details actually*  
011 *occurred). Evaluating on a 100-video benchmark across six*  
012 *diverse domains, we compare whole-window LLM reasoning*  
013 *(Llama-3.1-8B), iterative cloze completion (Phi-3-mini-4k),*  
014 *and non-parametric visual vector interpolation across multi-*  
015 *second masked gaps ( $W = 3$ ). Our empirical analysis re-*  
016 *veals two fundamental phenomena: (1) **The Modality Decou-***  
017 ***pling Law**: while visual vector continuity collapses catas-*  
018 *trophically under optical motion variance ( $r = -0.865$ ,  $p <$*   
019  *$0.0001$ ), LLM narrative deduction is statistically invariant to*  
020 *camera and optical dynamics ( $r = -0.001$ ,  $p = 0.99$ ). (2)*  
021 ***Scale-Induced Surprisal Immunity**: smaller SLMs (3.8B)*  
022 *suffer steep performance degradation under high a priori*  
023 *caption surprisal ( $r = -0.641$ ,  $p < 0.0001$ ), whereas 8B-*  
024 *scale parametric models exhibit complete statistical immu-*  
025 *nity ( $r = +0.028$ ,  $p = 0.79$ ), sustaining an overall Mean*  
026 *Reciprocal Rank (MRR) of 0.558 (+26.5% relative gain over*  
027 *Phi-3). These findings demonstrate that symbolic narrative*  
028 *logic reliably bridges visual blind spots, offering concrete*  
029 *mathematical criteria for dynamic token pruning and tempo-*  
030 *ral compression in next-generation multimodal architectures.*  
031

## 032 1. Introduction

033 Modern Video-Language Models (Video-LLMs) [4, 5] have  
034 achieved impressive capabilities in video question answer-  
035 ing and temporal grounding. However, their computational

architecture relies almost exclusively on dense, brute-force  
visual encoding: high-rate video frames are projected into  
continuous embedding spaces and consumed uniformly by  
transformer backbones. This design overlooks a fundamental  
property of human physical reality: *narrative predictability*.

Consider a cooking sequence where an instructor slices  
vegetables and later plates a cooked stir-fry. Even if interme-  
diate camera footage is missing or blurred, any competent ob-  
server—and indeed, any large language model with sufficient  
world knowledge—can deduce with high confidence that  
the chef heated the wok, sauteed the aromatics, and added  
the vegetables. The precise visual pixels (such as optical  
flare or hand tremor) are *stochastic residuals*, but the *causal*  
*narrative* is procedurally deterministic. Conversely, in un-  
constrained wildlife footage, a predator’s sudden change in  
direction cannot be deduced from prior trajectory alone and  
demands continuous optical observation.

To rigorously interrogate this trade-off, we formalize the  
task of **Dense Caption Reconstruction**. Given a sequence  
of dense semantic events where an entire temporal window of  
consecutive intervals is masked out, we evaluate whether lan-  
guage models can reconstruct the missing narrative sequence  
purely from temporal context, and compare this against non-  
parametric visual vector continuity.

Through systematic cross-evaluation of prior informa-  
tional metrics and post-reconstruction performance across  
100 wild videos, this work delivers three primary contribu-  
tions:

- **Modality Decoupling Law**: We prove mathematically and empirically that visual continuity and symbolic language reasoning are completely orthogonal. Visual interpolation collapses under optical motion variance ( $r = -0.865$ ), whereas language models reason causally regardless of camera dynamics ( $r = -0.001$ ).
- **Surprisal Immunity at Scale**: We demonstrate that while compact 3.8B SLMs collapse when scene descriptions deviate from stereotypical phrases ( $r = -0.641$ ), 8B models possess the parametric world knowledge to reconstruct high-entropy events without degradation ( $r = +0.028$ ).
- **Diagnostic Bottleneck Metrics**: We show that minimum

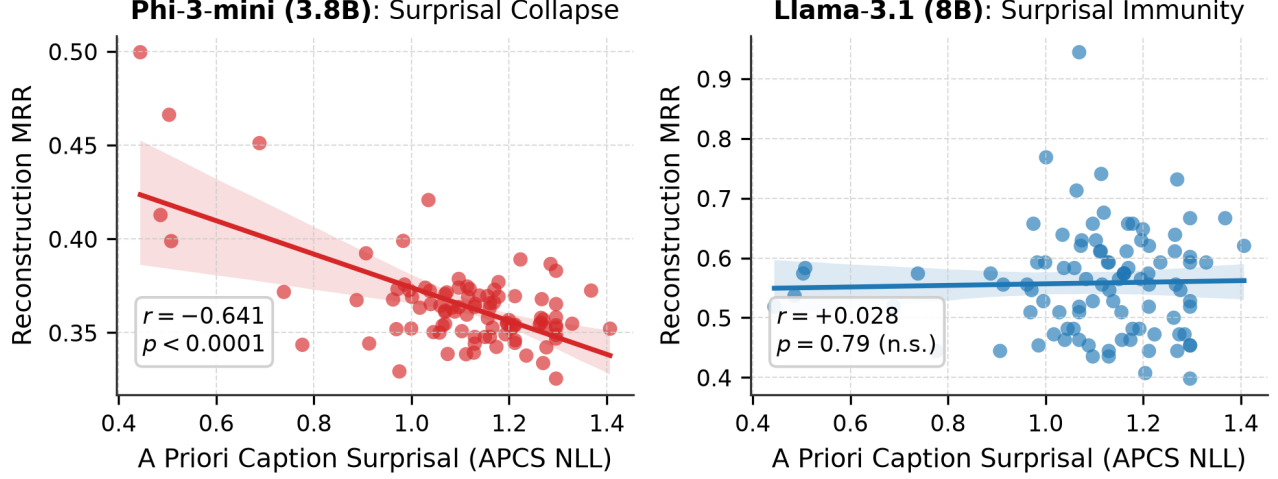


Figure 1. **Scale-Induced Surprisal Immunity.** Under increasing a priori caption surprisal (APCS NLL), the smaller 3.8B model (Phi-3-mini) collapses ( $r = -0.641$ ,  $p < 0.0001$ ), whereas the 8B model (Llama-3.1) exhibits complete statistical immunity ( $r = +0.028$ ,  $p = 0.79$ ), proving that larger parametric capacity is essential for out-of-distribution causal reasoning.

cosine similarity ( $\cos \text{Sim}_{\min}$ ) serves as an acute indicator of localized temporal rupture, and uncover that backward causal deduction makes opening video sequences ( $i = 0$ ) the most accurately reconstructible positions on the benchmark.

## 2. Methodology

### 2.1. Problem Formulation

Let a video  $V$  be represented as an ordered sequence of  $T$  synchronized semantic intervals:

$$V = \{(v_t, c_t)\}_{t=1}^T \quad (1)$$

where  $v_t \in \mathbb{R}^D$  denotes the visual embedding of interval  $t$  (extracted via CLIP-ViT [6]), and  $c_t$  denotes its dense natural language caption. We define a masked temporal window  $M \subset \{1, \dots, T\}$  of width  $W = |M|$ . The observed context is  $O = \{t \in \{1, \dots, T\} \mid t \notin M\}$ . The objective is to reconstruct the semantic embeddings of the unobserved intervals  $\{\hat{e}_t\}_{t \in M}$ .

### 2.2. Reconstruction Pathways

**Whole-Window Parametric Reasoning:** We condition an instruction-tuned LLM (Llama-3.1-8B-Instruct [3]) on the complete observed caption history  $\{c_t\}_{t \in O}$  with explicit temporal gap markers indicating the timestamp span. The model generates the entire sequence of missing captions  $\{\hat{c}_t\}_{t \in M}$  simultaneously. Reconstructed captions are then mapped to semantic embeddings via a frozen text encoder:  $\hat{e}_{\text{text}}(t) = \mathcal{E}_{\text{text}}(\hat{c}_t)$ .

**Iterative Cloze Baseline:** As an SLM baseline, we evaluate Phi-3-mini-4k (3.8B) [1] operating in an iterative fill-in-the-

blank regime, predicting each missing segment sequentially using immediate neighboring context.

**Non-Parametric Visual Continuity:** To test whether visual motion interpolation suffices without symbolic understanding, we compute  $\hat{e}_{\text{vis}}(t)$  via temporal linear and nearest-boundary interpolation of observed visual vectors  $\{v_t\}_{t \in O}$ .

### 2.3. A Priori Informational Metrics

To diagnose when and why reconstruction succeeds, we establish two modalities of a priori properties:

**A Priori Caption Surprisal (APCS):** We compute the average cross-entropy negative log-likelihood (NLL) and perplexity of the ground truth captions under a causal language model:

$$\text{APCS}_{\text{NLL}} = -\frac{1}{N} \sum_{i=1}^N \log P(w_i \mid w_{<i}) \quad (2)$$

High APCS indicates rare vocabulary, complex phrasing, or unexpected semantic transitions.

**Visual Optical Motion Dynamics:** We quantify optical instability between consecutive video frame embeddings:

$$\Delta v_t = \|v_t - v_{t-1}\|_2, \quad \sigma_{\Delta v}^2 = \text{Var}(\{\Delta v_t\}_{t=2}^T) \quad (3)$$

where  $\sigma_{\Delta v}^2$  captures visual motion variance, abrupt cuts, and high-frequency camera transitions.

### 2.4. Post-Reconstruction Evaluation

For each missing segment  $t \in M$ , we evaluate candidate retrieval over the universe of video segments  $\mathcal{U}$ . The model ranks candidates by cosine similarity:

$$\text{Sim}(e_1, e_2) = \frac{e_1 \cdot e_2}{\|e_1\| \|e_2\|} \quad (4)$$

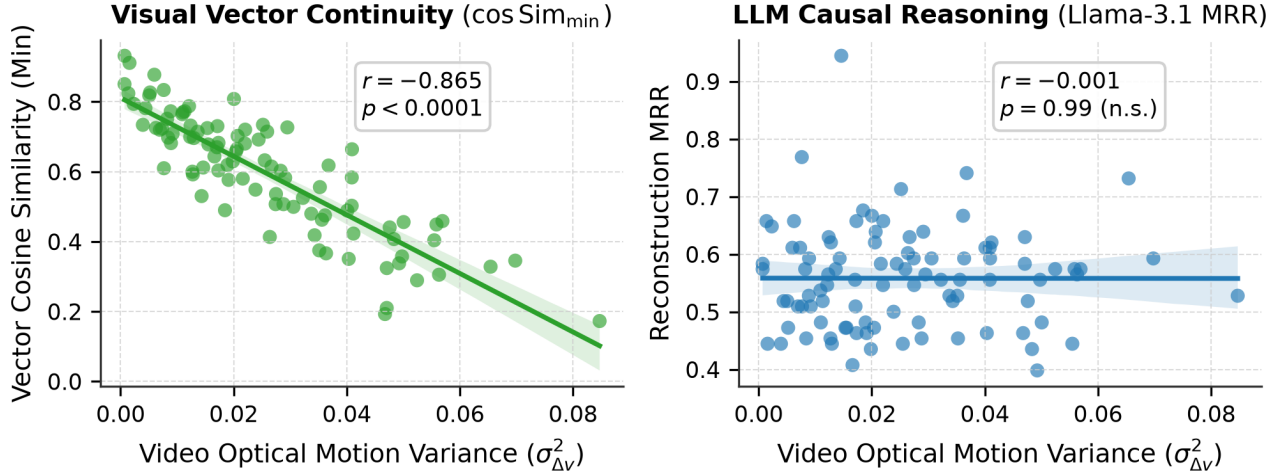


Figure 2. **The Modality Decoupling Law.** (Left) Visual vector continuity degrades precipitously as optical motion variance ( $\sigma_{\Delta v}^2$ ) increases ( $r = -0.865, p < 0.0001$ ). (Right) LLM causal reasoning (Llama-3.1 MRR) is completely invariant to optical motion dynamics ( $r = -0.001, p = 0.99$ ), establishing that language models deduce narrative progress independently of camera shake.

Table 1. **Statistical Correlation Matrix** between a priori properties and post-reconstruction metrics across 95 fully cross-evaluated videos. Pearson  $r$  and Spearman  $\rho$  (\*:  $p < 0.05$ , \*\*:  $p < 0.01$ , \*\*\*:  $p < 0.001$ , ns: not significant).

| Prior Metric                     | Post Metric                    | Pearson $r$    | Sig. | Spearman $\rho$ | Sig. |
|----------------------------------|--------------------------------|----------------|------|-----------------|------|
| APCS NLL                         | Phi-3 MRR                      | <b>-0.6408</b> | ***  | -0.3405         | ***  |
| APCS NLL                         | Llama-3.1 MRR                  | <b>+0.0276</b> | ns   | +0.0365         | ns   |
| APCS NLL                         | Llama vs. Phi $\Delta$         | <b>+0.2022</b> | *    | +0.1446         | ns   |
| APCS NLL                         | Visual MRR                     | -0.1417        | ns   | -0.2261         | *    |
| Motion Avg                       | Visual cos Sim <sub>mean</sub> | <b>-0.8613</b> | ***  | -0.8617         | ***  |
| Motion Avg                       | Visual cos Sim <sub>min</sub>  | <b>-0.8264</b> | ***  | -0.8293         | ***  |
| Motion Avg                       | Llama-3.1 MRR                  | -0.0808        | ns   | -0.0813         | ns   |
| Motion Var $\sigma_{\Delta v}^2$ | Visual cos Sim <sub>min</sub>  | <b>-0.8648</b> | ***  | -0.8622         | ***  |
| Motion Var $\sigma_{\Delta v}^2$ | Llama cos Sim <sub>min</sub>   | <b>-0.3106</b> | **   | -0.3087         | **   |
| Motion Var $\sigma_{\Delta v}^2$ | Llama-3.1 MRR                  | <b>-0.0013</b> | ns   | +0.0304         | ns   |
| Motion Var $\sigma_{\Delta v}^2$ | Phi-3 MRR                      | +0.0290        | ns   | -0.0748         | ns   |

We report **Mean Reciprocal Rank (MRR)** ( $\frac{1}{|U|} \sum \frac{1}{\text{Rank}}$ ), **Recall@1**, and both **Mean** and **Minimum Cosine Similarity** ( $\text{cos Sim}_{\min} = \min_{t \in M} \text{Sim}(\hat{e}_t, e_t^*)$ ).

### 3. Experiments and Results

#### 3.1. Benchmark Dataset

We evaluate on 100 wild videos sampled from the WildQA corpus [2], spanning six diverse categories: *Farming*, *Survival*, *Military*, *Action/Vehicle*, *Nature/Documentary*, and *Scenery*. Each video contains dense 3-second interval annotations. We evaluate across a continuous masking width of  $W = 3$  (a 9-second narrative omission), with evaluation instances placed across Opening ( $i = 0$ ), Middle, and Ending segments, yielding 300 reconstruction episodes.

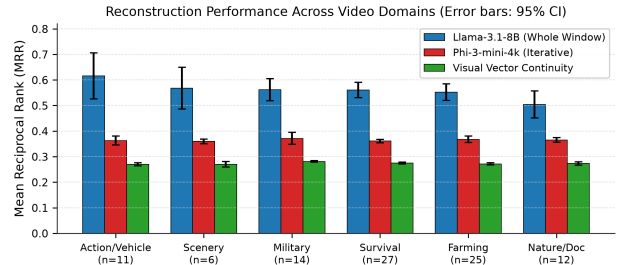


Figure 3. **Reconstruction Across Domains.** Error bars denote two-tailed 95% Student’s  $t$  confidence intervals. While between-category differences for Llama-3.1 are not statistically significant ( $p = 0.096$ ), its paired advantage over Visual Continuity and Phi-3 is consistent across all domains ( $p < 0.0001$ ).

#### 3.2. Finding 1: Scale-Induced Surprisal Immunity

As presented in Figure 1 and Table 1, prior caption surprisal (APCS NLL) exhibits a massive negative correlation with Phi-3-mini performance ( $r = -0.6408, p < 0.0001$ , power = 100%). Compact models lean heavily on high-probability stereotypical phrases; when real-world events diverge from standard templates, their generative capability collapses. In sharp contrast, Llama-3.1-8B shows no significant correlation with APCS NLL ( $r = +0.0276, p = 0.7908$ ). Comparing these two dependent correlations via Williams’ test reveals that Llama’s robustness shift relative to Phi-3 is statistically decisive ( $t = -5.51, p = 3.2 \times 10^{-7}$ ). Consequently, Llama’s margin of victory over Phi-3 increases significantly as caption surprisal rises ( $r = +0.2022, p = 0.0495$ ).

#### 3.3. Finding 2: The Modality Decoupling Law

Figure 2 empirically validates the Modality Decoupling Law. For non-parametric visual interpolation, motion variance acts as an immediate failure mode: visual cosine sim-

Table 2. **Exploratory Domain Breakdown.** Performance across video categories ( $W = 3$ ). Between-category variation in Llama-3.1 MRR is not statistically significant (ANOVA  $F = 1.94, p = 0.096$ ), but within-category paired gains over Visual Continuity are massive ( $p < 0.0001$ ).

| Domain              | N         | APCS         | Motion       | Phi-3        | Visual       | Llama-3.1    |
|---------------------|-----------|--------------|--------------|--------------|--------------|--------------|
| Action/Vehicle      | 11        | 1.168        | 0.146        | 0.363        | 0.270        | <b>0.615</b> |
| Scenery             | 6         | 1.190        | 0.085        | 0.360        | 0.270        | <b>0.568</b> |
| Military            | 14        | 1.112        | 0.194        | 0.371        | 0.281        | <b>0.562</b> |
| Survival            | 27        | 1.073        | 0.150        | 0.361        | 0.275        | <b>0.560</b> |
| Farming             | 25        | 1.024        | 0.196        | 0.368        | 0.272        | <b>0.552</b> |
| Nature/Doc          | 12        | 1.223        | 0.190        | 0.365        | 0.273        | <b>0.505</b> |
| <b>Overall Mean</b> | <b>95</b> | <b>1.103</b> | <b>0.165</b> | <b>0.365</b> | <b>0.274</b> | <b>0.558</b> |

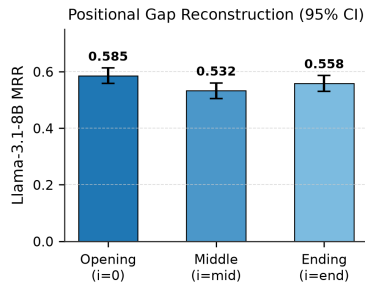


Figure 4. **Positional Reconstruction MRR (95% CI).** Opening sequences ( $i = 0$ ) outperform middle segments ( $p = 0.0058$ ), reflecting backward causal constraints in edited video.

ilarity degrades dramatically with motion variance ( $r = -0.8648, p < 0.0001$ , power = 100%). When a scene undergoes rapid panning, cutting, or erratic subject displacement, visual interpolation fails because the optical manifold breaks temporal smoothness.

Language reasoning, however, is decoupled from visual motion: the correlation between optical motion variance and Llama-3.1 MRR is virtually zero ( $r = -0.0013, p = 0.9899$ ). A textual description (e.g., “*The combine harvester turns 90 degrees at the edge of the crop row*”) remains fully deducible regardless of whether the physical recording camera shook or hovered smoothly.

### 3.4. Finding 3: Bottleneck Metric and Positional Teleology

**Sensitivity of  $\cos \text{Sim}_{\min}$ :** As shown in Table 1, minimum cosine similarity exhibits stronger sensitivity to motion shocks ( $r = -0.3106, p = 0.0022$ ) than mean similarity ( $r = -0.2568, p = 0.0120$ ). Mean similarity masks severe localized ruptures in multi-second gaps, whereas  $\cos \text{Sim}_{\min}$  accurately flags the single weakest link.

**Positional Teleology:** Reconstructing opening sequences ( $i = 0$ ) achieves significantly higher retrieval MRR ( $0.585 \pm 0.028$ ) than middle segments ( $0.532 \pm 0.028$ , paired  $t = 2.82, p = 0.0058$ ), while the difference between opening and ending segments ( $0.558 \pm 0.028$ ) is not statistically significant ( $p = 0.177$ ) (Figure 4). In natural video editing, downstream narrative resolution is strongly determined by

initial conditions; working backward from the established premise provides tighter causal constraints than forward open-ended branching.

## 4. Discussion and Implications

**Dynamic Multimodal Token Compression:** Our results offer a concrete architectural design guideline for Video-LLMs: rather than allocating uniform frame tokens across time, architectures should compute local motion variance  $\sigma_{\Delta v}^2$  and caption predictability. In quiescent, low-variance intervals, visual embeddings can be compressed via simple temporal interpolation. Conversely, when visual surprisal spikes or frames are intentionally dropped for efficiency, the system should rely on symbolic language reasoning.

**The Event Horizon:** While an 8B model successfully bridges 9-second gaps ( $W = 3$ ) with 0.558 MRR, future work must establish the critical gap threshold—the “Event Horizon” ( $W \geq 6$ )—where narrative divergence turns chaotic and parametric world models can no longer constrain physical reality.

## 5. Conclusion

Through dense caption reconstruction, we demonstrated that language models operate as robust causal engines capable of bridging temporal omissions in video. We established that parametric scale ( $\geq 8B$ ) eliminates vulnerability to language surprisal, and proved that symbolic reasoning is mathematically decoupled from optical motion disturbance. These principles pave the way toward selective, semantically guided video processing architectures.

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